MF 281: Probability Theory

Week 3: Conditional probability and Bayes' rule

What a probability becomes once you know something

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Lecture notes for students of MF 281 Probability Theory.

Week 3: Conditional probability and Bayes' rule

Learning Objectives

By the end of this week you should be able to:

- define conditional probability, and show that it is itself a probability law;
- use the multiplication and chain rules, with or without a tree diagram;
- apply the total probability theorem over a partition;
- state and apply Bayes' rule, and name the prior, the likelihood and the posterior;
- explain why a test that is right 95% of the time can still be wrong most times it fires;
- avoid the two standard errors: confusing $P(A \mid B)$ with $P(B \mid A)$, and ignoring the base rate.

3.1 Conditioning

Learning something changes what you should believe. If I tell you the die showed an even number, the chance it showed a 2 is no longer $1/6$ but $1/3$: the sample space has shrunk from six outcomes to three, and the 2 is one of them.

Definition 3.1 (Conditional probability)**.** For events A and B with $P(B) > 0$,

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

The denominator renormalises: we are no longer measuring A against the whole of Ω, but against the part of it where B happened. Nothing is defined when $P(B) = 0$, because dividing the measure of nothing by nothing says nothing.

Proposition 3.1. *For fixed B with $P(B) > 0$, the function $A \mapsto P(A \mid B)$ satisfies the three axioms, and so is a probability law in its own right.*

Proof. Non-negativity is immediate, since both numerator and denominator are non-negative. Normalisation: $P(\Omega \mid B) = P(\Omega \cap B)/P(B) = P(B)/P(B) = 1$. For countable additivity, let $A_1, A_2, \ldots$ be disjoint. The sets $A_i \cap B$ are then also disjoint, so

$$P\left(\bigcup_i A_i \,\middle|\, B\right) = \frac{P(\bigcup_i (A_i \cap B))}{P(B)} = \frac{\sum_i P(A_i \cap B)}{P(B)} = \sum_i P(A_i \mid B). \qquad \square$$

That proposition is the licence to do everything twice. Every identity proved for P, the addition rule included, holds for $P(\,\cdot \mid B)$ as well. Conditioning does not create a new theory; it restricts the old one to a smaller world.

Example 1: a die, conditioned

Roll a fair die. Let $A = \{2\}$ and $B = \{2, 4, 6\}$.

Solution

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{1/6}{1/2} = \frac{1}{3}.$$

Going the other way, $P(B \mid A) = \dfrac{1/6}{1/6} = 1$: if the die shows 2 it is certainly even. The two conditionals are $1/3$ and 1, and confusing them is the commonest error in this chapter.

3.2 The multiplication and chain rules

Rearranging the definition gives

$$P(A \cap B) = P(B)\,P(A \mid B) = P(A)\,P(B \mid A),$$

and extending it,

$$P(A_1 \cap A_2 \cap \cdots \cap A_n) = P(A_1)\,P(A_2 \mid A_1)\,P(A_3 \mid A_1 \cap A_2) \cdots P(A_n \mid A_1 \cap \cdots \cap A_{n-1}).$$

Example 2: three hearts

Draw three cards from a standard deck without replacement. What is the probability all three are hearts?

Solution

By the chain rule, conditioning on what has already gone:

$$\frac{13}{52} \times \frac{12}{51} \times \frac{11}{50} = \frac{11}{850} = 0.0129.$$

Each factor is a conditional probability: after one heart has gone, 12 of the remaining 51 cards are hearts. Sampling without replacement is exactly where the chain rule earns its keep.

3.2.1 Tree diagrams

A tree puts the chain rule on paper. Branches from a node are the possibilities given everything upstream; multiply along a path, add across paths.

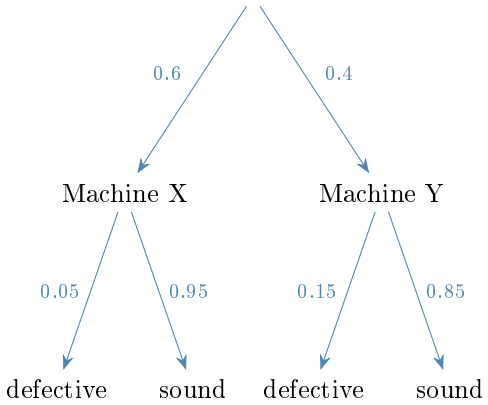


Figure 3.1: Multiply along a path: a defective item from X has probability $0.6 \times 0.05 = 0.03$. Add across paths for the total.

3.3 The total probability theorem

Theorem 3.2 (Total probability)**.** *If $B_1, \ldots, B_n$ partition Ω, each with $P(B_i) > 0$, then for any event A*

$$P(A) = \sum_{i=1}^{n} P(B_i)\,P(A \mid B_i).$$

Proof. The sets $A \cap B_i$ are disjoint and their union is A, so by finite additivity $P(A) = \sum_i P(A \cap B_i)$, and each term equals $P(B_i)P(A \mid B_i)$ by the multiplication rule. □

In words: split into cases, weight each case by how likely it is, and add. The cases must partition, which is why Week 1 spent time on partitions.

Example 3: two machines

A factory makes an item on machine X (60% of output, 5% defective) or machine Y (40%, 15% defective). An item is picked at random.

Solution

$$P(D) = 0.6(0.05) + 0.4(0.15) = 0.03 + 0.06 = 0.09.$$

Nine percent defective overall, which sits between the two rates and nearer to X's because X makes more. A quick sanity check on any total probability calculation: the answer must lie between the smallest and largest conditional probability.

3.4 Bayes' rule

Example 3 answered: given the machine, how likely is a defect? The useful question is usually the reverse: given a defect, which machine?

Theorem 3.3 (Bayes' rule)**.** *For a partition $B_1, \dots, B_n$ and an event A with $P(A) > 0$,*

$$P(B_k \mid A) = \frac{P(B_k)\,P(A \mid B_k)}{\sum_{i=1}^{n} P(B_i)\,P(A \mid B_i)}.$$

Proof. $P(B_k \mid A) = P(A \cap B_k)/P(A)$. The numerator is $P(B_k)P(A \mid B_k)$ by the multiplication rule, and the denominator is the total probability theorem. □

The three pieces have names worth learning: $P(B_k)$ is the **prior**, what you believed before; $P(A \mid B_k)$ is the **likelihood**, how well each cause explains what you saw; and $P(B_k \mid A)$ is the **posterior**, what you believe afterwards.

Example 4: which machine made it

Continuing Example 3, a defective item is found. What is the probability it came from machine Y?

Solution

$$P(Y \mid D) = \frac{0.4(0.15)}{0.09} = \frac{0.06}{0.09} = \frac{2}{3} = 0.667.$$

Machine Y makes only 40% of the output but two thirds of the defects: the defect is evidence about the source. Note also $P(X \mid D) = 0.03/0.09 = 1/3$, and the two posteriors sum to one, as they must.

3.5 Why a good test gives bad news

Example 5: the false positive problem

A sensor detects a fault with probability 0.90 when a fault is present, and raises a false alarm with probability 0.05 when none is. One percent of units are faulty. The sensor fires. What is the probability the unit is actually faulty?

Solution

With F for faulty and $+$ for an alarm,

$$P(F \mid +) = \frac{0.01(0.90)}{0.01(0.90) + 0.99(0.05)} = \frac{0.0090}{0.0090 + 0.0495} = \frac{0.0090}{0.0585} = 0.154.$$

Fewer than one alarm in six is real.

The same thing in whole units

Take 10,000 units. 100 are faulty, of which the sensor catches 90. Of the 9900 sound ones, it wrongly flags 495. So 585 alarms, of which 90 are genuine: $90/585 = 15.4\%$.

Most people find the second version obvious and the first surprising, which is a useful teaching fact: when a conditional probability argument meets resistance, translate it into counts out of ten thousand.

What drives it

The base rate. Sound units outnumber faulty ones 99 to 1, so even a small false alarm rate produces more false alarms than true ones. Raise the prevalence to 10% and the same sensor gives $P(F \mid +) = 0.667$. The test did not change; the population did.

$P(+ \mid F) = 0.90$ and $P(F \mid +) = 0.15$ are both true of the same sensor. Treating them as the same number is called the *prosecutor's fallacy*, and it has sent people to prison. When you read any claim about a test, ask which conditional is being quoted.

Example 6: three suppliers

Bolts come from three suppliers: 50% from S1 (2% defective), 30% from S2 (3%), 20% from S3 (5%). A defective bolt is found. Which supplier is most likely?

Solution

Total probability first:

$$P(D) = 0.5(0.02) + 0.3(0.03) + 0.2(0.05) = 0.010 + 0.009 + 0.010 = 0.029.$$

Then the posteriors:

$$P(S_1 \mid D) = \frac{0.010}{0.029} = 0.345, \quad P(S_2 \mid D) = \frac{0.009}{0.029} = 0.310, \quad P(S_3 \mid D) = \frac{0.010}{0.029} = 0.345.$$

S1 and S3 are equally likely, although S3's defect rate is more than twice S1's: S1 supplies two and a half times as many bolts. Blaming the supplier with the worst rate would be wrong half the time here.

3.6 A famous one

Example 7: the three doors

Three doors hide one prize and two goats. You pick door 1. The host, who knows where the prize is, opens door 3 to reveal a goat and offers you the switch. Should you?

Solution

Let the prize be behind door i with prior $1/3$ each, and let H_3 be the event that the host opens door 3. The likelihoods are the crux:

$$P(H_3 \mid \text{prize behind } 1) = \tfrac{1}{2}, \quad P(H_3 \mid \text{prize behind } 2) = 1, \quad P(H_3 \mid \text{prize behind } 3) = 0.$$

The host must avoid the prize and your door, so with the prize behind 2 he has no choice. Then

$$P(H_3) = \tfrac{1}{3}\left(\tfrac{1}{2}\right) + \tfrac{1}{3}(1) + \tfrac{1}{3}(0) = \tfrac{1}{2},$$

$$P(\text{behind } 1 \mid H_3) = \frac{\frac{1}{3}\cdot\frac{1}{2}}{\frac{1}{2}} = \tfrac{1}{3}, \qquad P(\text{behind } 2 \mid H_3) = \frac{\frac{1}{3}\cdot 1}{\frac{1}{2}} = \tfrac{2}{3}.$$

Switch: it doubles your chance.

Why it feels wrong

Because the host's choice is not random. His action carries information precisely because he was constrained by where the prize is. Change the story so that the host opens a door at random and happens to reveal a goat, and the two doors become equally likely. The likelihoods, not the counting, do all the work.

3.7 Two errors to avoid

1. **Reversing the conditional.** $P(\text{alarm} \mid \text{fault})$ is a property of the sensor; $P(\text{fault} \mid \text{alarm})$ is what you want, and they differ by the base rate.

2. **Forgetting the prior.** A rare condition stays unlikely after weak evidence. Any argument that reaches a confident conclusion without using the prior has smuggled one in.

3.8 Summary

Week 3 in six lines

- $P(A \mid B) = P(A \cap B)/P(B)$, and it is a probability law in its own right.
- The chain rule handles sampling without replacement, one draw at a time.
- Total probability: split into cases that partition, weight, add.
- Bayes turns likelihoods into posteriors, using the prior.
- A 90% sensor on a 1% problem is right about one alarm in six.
- $P(A \mid B)$ and $P(B \mid A)$ are different numbers.

Exercises

1. $P(A \cap B) = 0.12$ and $P(B) = 0.3$. Find $P(A \mid B)$. If also $P(A) = 0.4$, find $P(B \mid A)$ and comment.

2. Prove that $P(A^c \mid B) = 1 - P(A \mid B)$ directly from the definition.

3. Two cards are drawn without replacement. Find the probability both are aces, and the probability the second is an ace given the first was.

4. A fair die is rolled four times. Find the probability of at least one six, and explain which rule from Week 2 you used.

5. A plant takes 70% of its ore from pit A (grade above cut-off 80% of the time) and 30% from pit B (60%). Find the probability a random load is above cut-off, and the probability it came from B given that it is.

6. A test is 99% sensitive and 98% specific for a condition affecting 1 in 1000. Find the probability of having the condition given a positive test, and express the calculation in whole people out of 100,000.

7. In Exercise 6, how high would the prevalence have to be for a positive test to mean a better than even chance of having the condition?

8. Three machines produce 25%, 35% and 40% of output with defect rates 1%, 2% and 3%. Find $P(\text{defective})$ and the three posteriors given a defect.

9. A student argues: "Most accidents happen near home, so driving near home is dangerous." Name the conditional being confused, and write both carefully.

10. Show that if $P(A \mid B) = P(A)$ then $P(B \mid A) = P(B)$, provided both probabilities are positive. What is this property called?

11. Repeat the three-door problem with four doors, where the host opens one goat door and offers the switch. What is the probability of winning by switching?

12. A sensor's manufacturer advertises "95% accurate". Explain why that phrase is not enough to compute anything, and state the two numbers you would ask for.

Answers to the odd-numbered exercises

1. $P(A \mid B) = 0.12/0.3 = 0.4$ and $P(B \mid A) = 0.12/0.4 = 0.3$. Here $P(A \mid B) = P(A)$ and $P(B \mid A) = P(B)$: the events are independent, which is next week's subject.

3. Both aces: $\frac{4}{52} \times \frac{3}{51} = \frac{1}{221} = 0.0045$. Second an ace given the first was: $3/51 = 1/17 = 0.0588$.

5. $P(\text{above}) = 0.7(0.8) + 0.3(0.6) = 0.56 + 0.18 = 0.74$, and $P(B \mid \text{above}) = 0.18/0.74 = 0.243$.

7. Writing p for prevalence, a positive test means better than even odds when $0.99p > 0.02(1-p)$, that is $p > 0.02/1.01 = 0.0198$, about 2%. Twenty times the original prevalence, for the same test.

9. The claim confuses $P(\text{near home} \mid \text{accident})$, which is high, with $P(\text{accident} \mid \text{near home})$, which is not. Most driving happens near home, so most of everything does.

11. With four doors your first pick is right with probability 1/4. After one goat door is opened, the remaining 3/4 is shared by the two unopened doors, so switching to a particular one wins with probability 3/8, which still beats 1/4.

Before next week

Attempt exercises 1 to 8, and bring Exercise 6 worked both ways, in probabilities and in whole people. Week 4 asks when conditioning changes nothing at all, which is the definition of independence, and shows why "independent in pairs" is weaker than it sounds.